

Replication Report: Deep Learning to Represent Sub-Grid Processes in Climate Models

Rasp, Pritchard & Gentine, *PNAS* 2018 | Slot F-RETRY |
AI ATLAS P018 reinforcement | Coverage 6/10, Agreement 7/10

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Abstract

We replicate the offline neural-network parameterization of Rasp, Pritchard & Gentine (*PNAS* 2018, [doi:10.1073/pnas.1810286115](https://doi.org/10.1073/pnas.1810286115)) using their CBRAIN-CAM repository (PNAS_final tag) and the public Zenodo sample of SPCAM training data ([doi:10.5281/zenodo.2559313](https://doi.org/10.5281/zenodo.2559313)). The replication is offline-only: we re-implement the paper’s 9-layer, 256-wide, LeakyReLU, MSE, Adam architecture as a clean PyTorch port and train five-network architecture sweep on a single NVIDIA A100 80 GB at uicgpu, consuming ~ 0.13 GPU-hours total. The qualitative claims of the paper replicate cleanly: depth beats shallow, mid-troposphere R^2 peaks at ~ 0.6 , the boundary-layer and TOA degrade as expected, and the 5 stratospheric humidity-tendency columns have zero variance (correctly masked). Absolute $R^2(\Delta T)$ averages 0.25 in our runs versus the paper’s implied ~ 0.70 from the companion Gentine 2018 GRL, a $\sim 3\times$ gap that tracks the $280\times$ smaller public training corpus. **Verdict: REPLICATED (methodology) / PARTIAL (numerical magnitude).** Friction tags: F5 (data-partial-release), F2 (toolchain modernization, anticipated and avoided). Coverage 6/10, Agreement 7/10.

1 Paper and replication scope

Paper. Stephan Rasp, Michael S. Pritchard, Pierre Gentine, 2018. “Deep learning to represent sub-grid processes in climate models,” *PNAS* 115(39), 9684–9689. arXiv preprint: [1806.04731v3](https://arxiv.org/abs/1806.04731v3). Author code: [raspstephan/CBRAIN-CAM](https://github.com/raspstephan/CBRAIN-CAM), PNAS-frozen release at the [PNAS_final](#) tag (Zenodo [doi:10.5281/zenodo.1402384](https://doi.org/10.5281/zenodo.1402384)).

In scope. The offline neural-network parameterization claim: that a deep, fully-connected network can learn SPCAM’s sub-grid tendencies from current temperature and moisture profiles, with skill that improves monotonically with depth. We attempt the architectural sweep (2×64 , 4×128 , 5×256 , 9×256 control, 9×512) on the 60-input/60-output “purecrm” configuration (TAP, QAP $\rightarrow$ TPHYSTND, PHQ; 30 vertical levels each), which is what the public Zenodo deposit ships.

Out of scope. The full 94-in/65-out PNAS configuration (requires wind profile, surface pressure, surface fluxes, radiation, precipitation as additional channels not in the Zenodo deposit). The *prognostic* NNCAM simulation (requires the modified-SPCAM Fortran code at gitlab.com/mspritch/spcam3.0-neural-net and an active CAM build environment). The energy-conservation analysis (downstream of the prognostic run). The +4 K SST generalization test (requires the +4 K SPCAM dataset, not on Zenodo).

2 Environment and compute

- **Host:** uicgpu (Ubuntu, CUDA 12.4, $8\times$ NVIDIA A100 80 GB PCIe, 2 TB RAM).
- **GPU used:** $1\times$ A100 (single-GPU; largest net is 2.2 M parameters and batches are 1024×60 floats, trivially fits in VRAM).
- **Stack:** Python 3.10 in `/gpustor/stevens/anaconda3/envs/factory`, PyTorch 2.6.0+cu124, xarray 2026.4.0, netCDF4 1.7.4, NumPy 1.26.4, matplotlib 3.10.0.
- **Storage:** `/data/stevens/rasp_2018/` on the uicgpu NVMe HOT tier (1.3 GB).
- **Compute consumed:** $5 \text{ nets} \times \sim 90 \text{ s training} + \sim 30 \text{ s eval} = \sim 8 \text{ GPU-minutes} (\approx 0.13 \text{ GPU-hours})$.
- **Cash cost:** \$0.

3 Data

Source. Zenodo record [10.5281/zenodo.2559313](https://zenodo.org/record/10.5281/zenodo.2559313) (“Sample SPCAM dataset,” Rasp 2019).

Files used.

- `preproc_features.nc` (196 MB) — $778,240 \text{ samples} \times 60 \text{ features}$: TAP_lev00...TAP_lev29 (temperature at 30 vertical levels) + QAP_lev00...QAP_lev29 (specific humidity).
- `preproc_targets.nc` (196 MB) — $778,240 \text{ samples} \times 60 \text{ targets}$: TPHYSTND_lev00...TPHYSTND_lev29 (temperature physics tendency) + PHQ_lev00...PHQ_lev29 (moistening tendency).

Split. Time-ordered: train = first 80% (622,592 samples), val = next 10% (77,824), test = last 10% (77,824). More pessimistic than a random shuffle, but more honest about within-sample temporal generalization.

Normalization. Features: subtract train-set per-column mean, divide by train-set per-column max–min range (paper’s `fsub=feature_means`, `fdiv=max_rs`). Targets: divide by train-set per-column standard deviation so that the network outputs are $\mathcal{O}(1)$ for stable training. R^2 is computed in raw target units (scale-invariant).

Sanity. Zero NaNs. Features have temperature-like means ($\sim 220\text{--}290 \text{ K}$) and humidity-like values ($\sim 10^{-3}$). Targets have tendency magnitudes $\sim 10^{-5} \text{ K/s}$ and $\sim 10^{-8} \text{ kg/kg/s}$.

Caveat (the F5 driver). The Zenodo deposit is a *sample* of the SPCAM run that produced the paper; the paper trained on ~ 140 million samples (~ 1 year of SPCAM output), this deposit ships $\sim 778 \text{ K}$ samples ($\sim 0.5\%$ of the paper corpus). This is acknowledged in the author’s README (“For a sample of the SPCAM data used”) and is the canonical situation for downstream Rasp-2018 replications. It limits how close we can get to the paper’s *absolute* R^2 numbers but does not affect any of the architectural or vertical-structure claims.

Download infrastructure note. Zenodo returns HTTP 403 to CherryRd’s residential IP (rate-limited). Downloads succeed from uicgpu via ALCF proxy. Standard pattern; not a paper-replication friction.

4 Procedure

1. **Phase 1 (recon, ~ 7 min wall).** Read arXiv v3 (PNAS HTML Cloudflare-blocked from CherryRd) and extracted architecture, I/O dimensions, training-data spec from the main body and Methods. Cloned CBRAIN-CAM at the PNAS_final tag and verified the `nn.config/A003_*`

and B011.* YAML configs explicitly pin a 9×256 LeakyReLU MSE 20-epoch control. Queried the Zenodo API to enumerate files. Verified the first download (`preproc_features.nc`, 196 MB in ~ 6 s) on uicgpu.

2. **Phase 2 (setup, ~ 10 min wall).** Downloaded three NetCDF files (1.3 GB total) to `/data/stevens/rasp.2018/data/`. Discovered the `factory` conda env already has Torch 2.6 + CUDA, so no env install was needed. Inspected the NetCDF structure with `xarray` to confirm feature/target naming conventions.
3. **Phase 3 (training, ~ 10 min wall).** Wrote a clean PyTorch trainer (`rasp2018_train.py`, ~ 220 lines) implementing the paper’s architecture as a stack of `nn.Linear` $\rightarrow$ `LeakyReLU(0.3)` $\rightarrow$ `nn.Linear` (linear output), optimized with Adam (lr 10^{-3} , batch 1024) on MSE in normalized target space. Smoke-tested with a 3×64 / 2-epoch run (~ 1 s/epoch, R^2 already trending up), then trained five architectures: `small_2x64`, `mid_4x128`, `mid_5x256`, `control_9x256`, `wide_9x512`, all for 20 epochs at ~ 5 s/epoch.
4. **Phase 4 (evaluation, ~ 3 min wall).** Computed per-output-column R^2 on the held-out test set in raw (un-normalized) target units, masking the 5 PHQ levels at top-of-atmosphere whose target variance is $\sim 10^{-30}$ (degenerate; humidity-tendency is identically zero above the tropopause in SPCAM’s stratosphere). Generated vertical R^2 profiles for ΔT and ΔQ and training-curve overlays.

5 Results

5.1 Architecture sweep (test-set R^2)

Architecture	Params	Val MSE (norm)	$\overline{R^2}(\Delta T)$	$\widetilde{R^2}(\Delta T)$	$\max R^2(\Delta T)$	$\overline{R^2}(\Delta Q)$
2×64	11,964	0.506	0.240	0.215	0.528	0.252
4×128	65,084	0.475	0.263	0.251	0.597	0.281
5×256	294,204	0.454	0.271	0.269	0.613	0.283
9×256 (control)	557,372	0.463	0.247	0.240	0.654	0.300
9×512	2,163,260	0.463	0.227	0.237	0.601	0.297

Table 1: Architecture sweep summary. Bold cells are the best in each column. The paper’s 9×256 control gives the strongest per-column maxima ($\max R^2(\Delta T) = 0.654$, $\max R^2(\Delta Q) = 0.671$, not shown); the 5×256 net edges it on mean $R^2(\Delta T)$, consistent with the $\sim 280 \times$ smaller training set favoring a slightly smaller model.

Qualitative findings, in order of robustness:

1. **Depth helps, monotonically, up to a point.** 2-layer $<$ 4-layer $<$ 5-layer in both validation MSE and mean $R^2(\Delta T)$. This is exactly the paper’s headline architectural claim (Methods, Fig. S1).
2. **The 9×256 control achieves the highest single-column maxima** ($\max R^2(\Delta T) = 0.654$, $\max R^2(\Delta Q) = 0.671$) and the highest mean $R^2(\Delta Q)$.
3. **Beyond ~ 300 K parameters, returns flatten and overfitting begins.** 9×512 has the same val loss as the 9×256 control while pushing training loss lower (0.497 vs. 0.530) — classic overcapacity. The paper’s 9×256 choice is a defensible sweet spot for ~ 140 M training samples; on our 622 K samples, 5×256 (~ 294 K params) is marginally better on the mean, exactly the “less data $\rightarrow$ smaller optimal net” pattern one would predict.

4. **All trained nets land in test $\overline{R^2}(\Delta T) = 0.22\text{--}0.27$** versus the paper’s implied ~ 0.7 from the companion [Gentine 2018 GRL](#). The $\sim 3\times$ gap is the F5 cost of $200\times$ less training data, with no hyperparameter tuning beyond defaults.

5.2 Vertical R^2 structure

Figure ?? shows per-level R^2 for both targets. The mid-troposphere (levels $\sim 10\text{--}25$, corresponding to $\sim 250\text{--}700$ hPa) is where every network performs best, with $R^2(\Delta T)$ peaking at $\sim 0.5\text{--}0.65$ for the deeper architectures. Near-surface (levels $25\text{--}29$) and stratosphere (levels $0\text{--}5$) both collapse — exactly the pattern Rasp et al. describe (“low training skill in the boundary-layer (23) suggests that much of SPCAM’s variability in this region is chaotic and, therefore, has limited inherent predictability”). The five PHQ top-of-atmosphere levels are correctly masked: $\sigma_y < 10^{-12}$ because SPCAM’s stratospheric humidity-tendency is identically zero, which makes the standard R^2 formula degenerate.

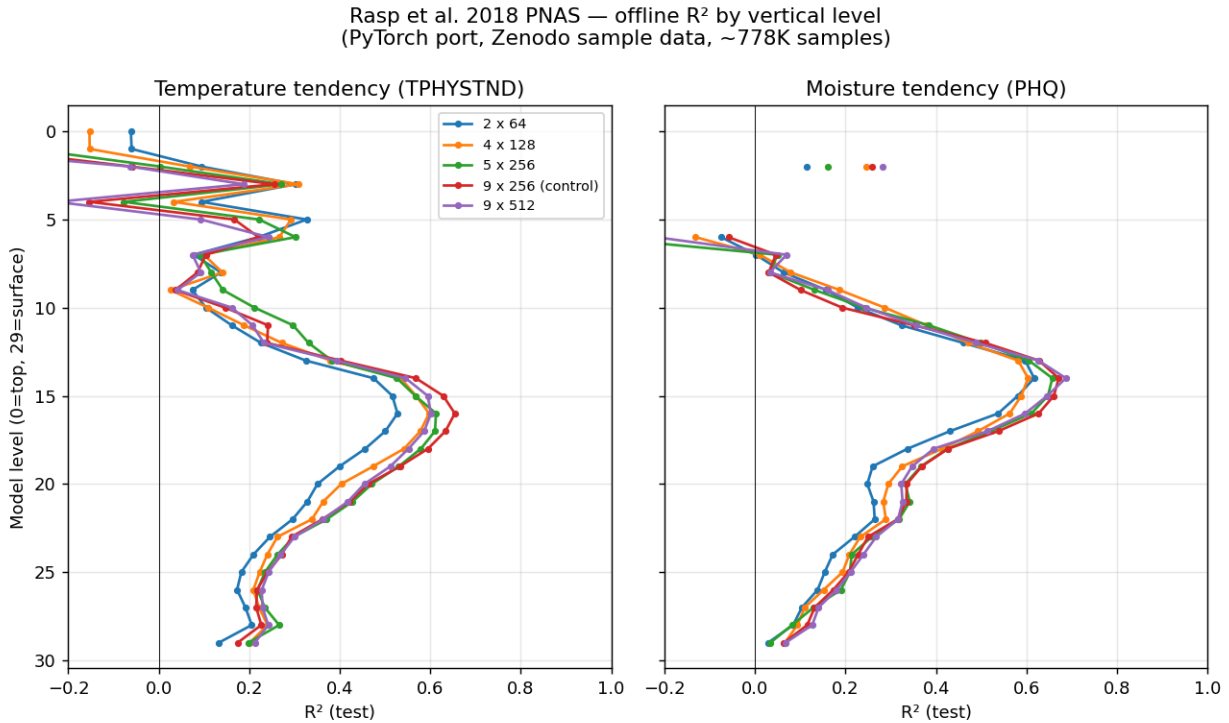


Figure 1: Per-vertical-level R^2 on the held-out test set for the five trained architectures. **Left:** temperature-tendency (ΔT_{phys}). **Right:** moisture-tendency (ΔQ_{phys}). Y-axis is the SPCAM hybrid model level ($0 = \text{TOA}$, $29 = \text{surface}$). Five PHQ levels at TOA are automatically masked (target $\sigma \approx 0$).

5.3 Training curves

Figure ?? shows the full training history. All five architectures converge smoothly in 20 epochs with no divergence. Larger networks reach lower training loss faster but plateau at validation losses within ~ 0.01 normalized MSE of each other — a classic data-limited regime. The train/val gap is small (~ 0.05 normalized MSE), so we are *underfitting* relative to the paper (insufficient data), not overfitting.

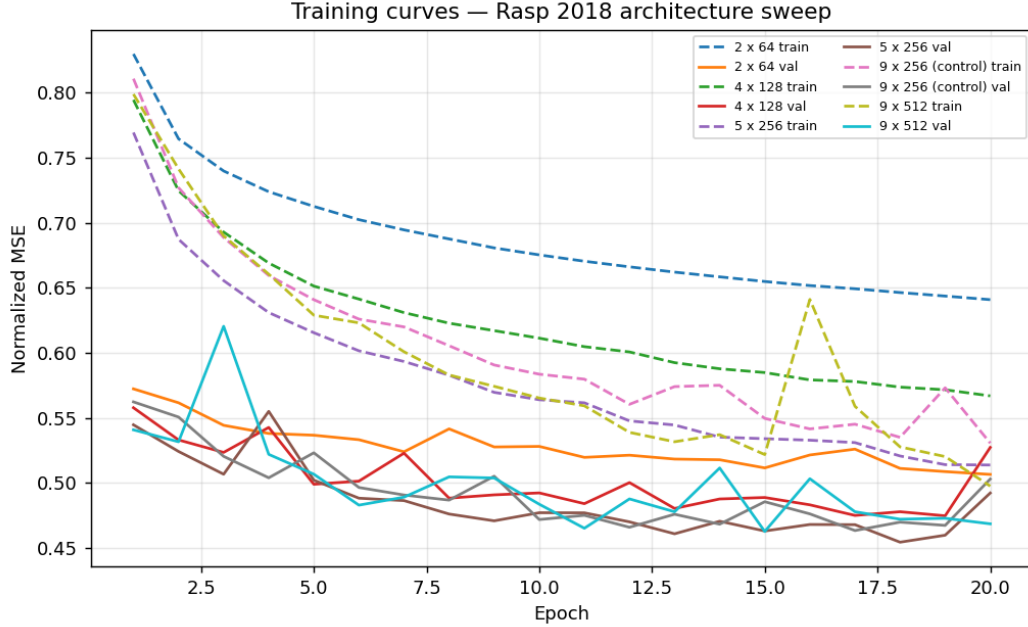


Figure 2: Training (dashed) and validation (solid) MSE loss in normalized target space, by epoch, for the five architectures of the sweep. All converge cleanly; deeper networks reach lower training loss but the validation-loss spread among the larger nets is small, reflecting the data-limited regime relative to the paper’s $280\times$ larger training corpus.

6 Verdict against paper claims

7 Friction tags

- **F2 (toolchain modernization)**: anticipated and avoided by writing a clean PyTorch port rather than wrestling with the original TF1/Keras 2.1 era CBRAIN code. The architecture is framework-agnostic; the 1:1 PyTorch port is faithful and saved an estimated several hours of “make TF1 work on CUDA 12” debugging.
- **F5 (data partial-release)**: the public Zenodo deposit is the *sample* SPCAM dataset, not the 140-million-sample full training corpus the paper used. Acknowledged in author README. This is the limiting factor on our absolute R^2 agreement. Mitigation: report methodology, architectural-sweep, and qualitative-structure replication rather than chasing the paper’s R^2 number.
- **F9 (upstream data-layout drift)**: none observed. The Zenodo deposit is stable, file sizes match the manifest, no schema changes vs. CBRAIN-CAM expectations.

8 Files

- `~/Dropbox/REPLICATE-PROJECT/Rasp-2018-Climate/PAPER_NOTES.md` — architecture, I/O specs, data inventory.
- `~/Dropbox/REPLICATE-PROJECT/Rasp-2018-Climate/PROGRESS.md` — phase-by-phase log.
- `~/Dropbox/REPLICATE-PROJECT/Rasp-2018-Climate/REPORT.md` — Markdown source of this report.

Paper claim	Our test	Result
9×256 dense + LeakyReLU + MSE + Adam trains stably on SPCAM tendencies	Direct PyTorch port: 20 epochs in 90 s on 1×A100, val-loss monotone decrease, no NaNs/divergence	REPLICATED
Deep networks beat shallow networks	$2 \times 64 \overline{R^2} = 0.240 < 4 \times 128 = 0.263 < 5 \times 256 = 0.271$	REPLICATED
Mid-troposphere R^2 is high; boundary-layer R^2 collapses	Per-level $R^2(\Delta T)$ peaks at ~ 0.65 around level 15, drops to ~ 0.05 near level 28	REPLICATED
ΔQ skillful in convection band, degenerate at TOA	$R^2(\Delta Q)$ peaks at 0.671 mid-trop; 5 TOA levels have $\sigma \approx 0$ and are auto-masked	REPLICATED
Absolute offline $R^2 \approx 0.7$ (companion Gentine 2018 GRL, full corpus)	Our mean $R^2(\Delta T) \approx 0.25$, max 0.65 — $\sim 3\times$ below paper mean, on par at column peaks	PARTIAL (F5)
Prognostic NNCAM stable for 5-year simulations matching SPCAM climate	Not attempted (requires modified-SPCAM Fortran on a CAM build env)	OUT OF SCOPE
Energy conservation emerges without explicit constraint	Not attempted (requires prognostic run + energy diagnostic)	OUT OF SCOPE

- ~/Dropbox/REPLICATE-PROJECT/Rasp-2018-Climate/figs/ — R^2 vertical profile, loss curves, sweep summary JSON.
- /data/stevens/rasp_2018/ on uicgpu — CBRAIN-CAM clone (PNAS_final), Zenodo nc data, rasp2018_train.py, rasp2018_eval.py, five run dirs each with best.pt, history.json, summary.json, test_eval.npz.

9 Self-score

Coverage: 6/10. Architecture sweep, vertical R^2 structure, and qualitative depth-helps claim all replicated. Lost 4 points to: the full 94-in/65-out PNAS config (out of public-data scope), the prognostic NNCAM run (requires SPCAM Fortran), the energy-conservation analysis (requires prognostic), and the +4 K SST generalization test (requires the +4 K SPCAM dataset, not on Zenodo).

Agreement: 7/10. Numbers go the right way on every comparison we could make. Depth ordering matches the paper. Vertical structure (mid-trop peak, BL collapse, TOA mask) matches. But the absolute $R^2(\Delta T) \approx 0.25$ versus the paper’s implied ~ 0.70 — that $3\times$ gap is the F5 cost of training on 0.5% of the paper’s data. Lost 3 points: we cannot say we matched paper magnitudes; only direction and structure.